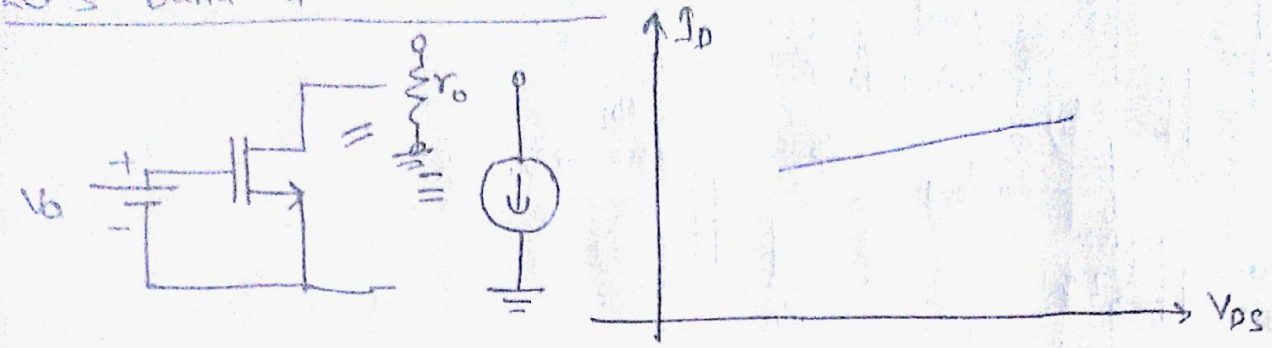
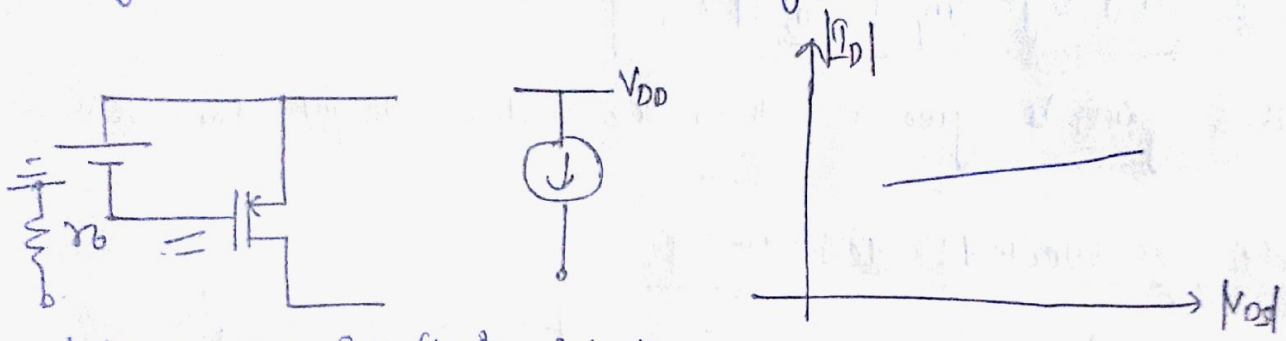
Lecture - 37

Let's build a current source! -

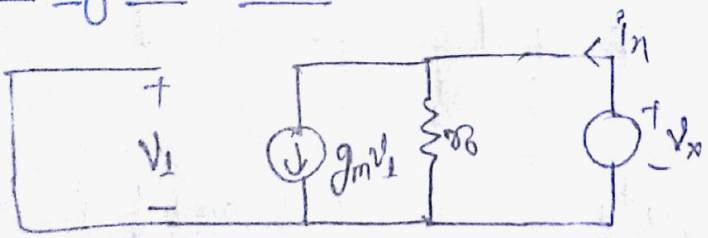


* In this current can start from drain but go to zero. It cannot go anywhere else. This current is flowing out of the node.



* In this current is flowing into the node.

Small signal model



$$\frac{V_x}{i_x} = r_o$$

* Any time MOS device acting as constant current source, we just replace it by r_o.

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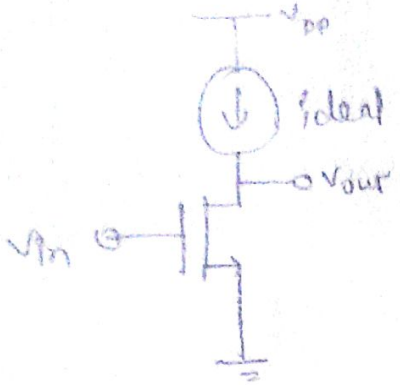
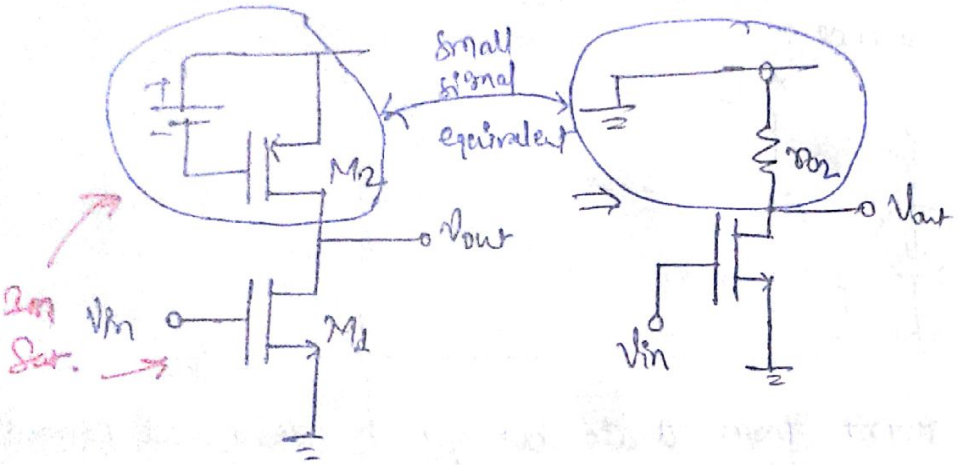
Common source stage with current source load:-



$\Rightarrow A_v = -g_m r_o$

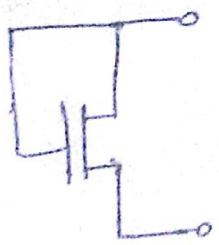
* In this current is pouring into the node so we PMOS device as current source.

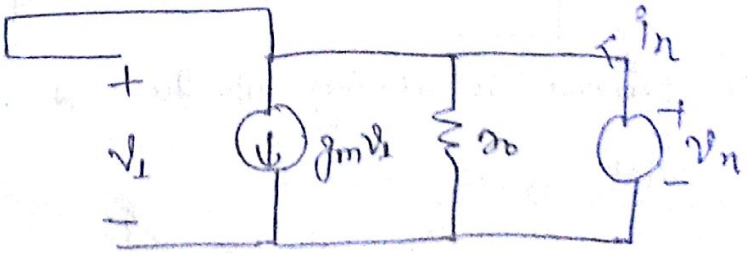


$$\boxed{A_v = -g_{m_1}\left(r_{o_2} \| r_{o_1}\right)}$$

* This gain is greater than the value we get for resistor only.

* Diode connected device:-





$V_1 = V_x$

$$i_x = \frac{V_x}{r_o} + g_m V_x \Rightarrow V_x\left(g_m + \frac{1}{r_o}\right) = i_x$$

$$\Rightarrow \frac{V_x}{i_x} = \frac{1}{\left(g_m + \frac{1}{r_o}\right)} = \left(r_o \| \frac{1}{g_m}\right)$$

It has low impedance level because of g_m.

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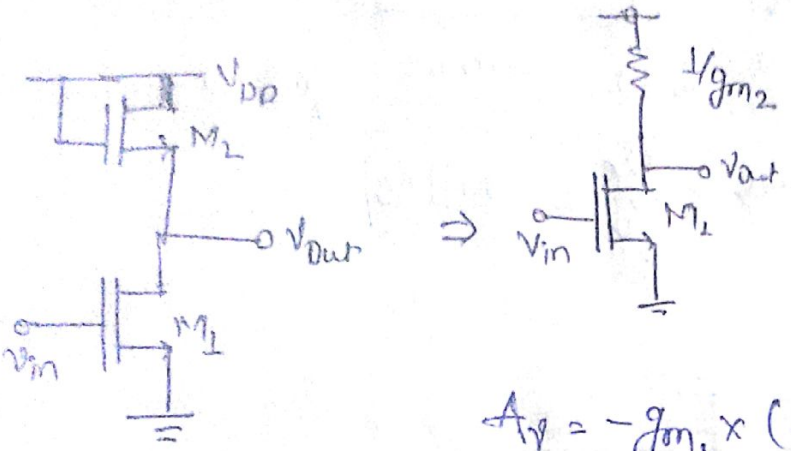
* For diode connected device, we just connect drain and gate.

* CS stage with diode connected load:-



$$A_v = -g_{m_1} \times \left(\frac{1}{g_{m_2}}\right)$$

$$= -\sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)_1 I_{D_1}} \times \frac{1}{\sqrt{2\mu_n C_{ox}\left(\frac{W}{L}\right)_2 I_{D_2}}}$$

$$\boxed{A_v = -\sqrt{\frac{(W/L)_1}{(W/L)_2}}} \quad \lambda = 0$$

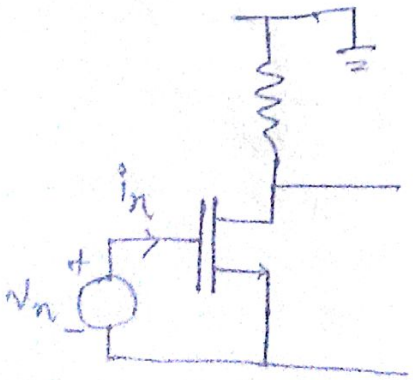
* $I_{D_1} = I_{D_2}$ because same current is flowing through the device

When $\lambda \neq 0$,

$$A_v = -g_{m_1}\left(\frac{1}{g_{m_2}} \| r_{o_1} \| r_{o_2}\right)$$

---

* Input and output impedances:-



$$\boxed{\text{Input Impedance} = \frac{V_x}{i_x} = \infty \text{ at low frequencies}}$$

Since i_x is gate current which is zero so input impedance becomes ∞. At low freq^n capacitance between gate and channel becomes open circuit.

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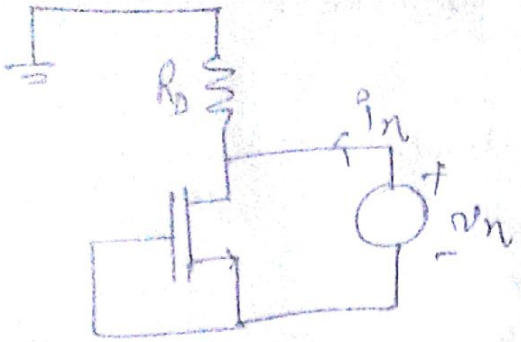
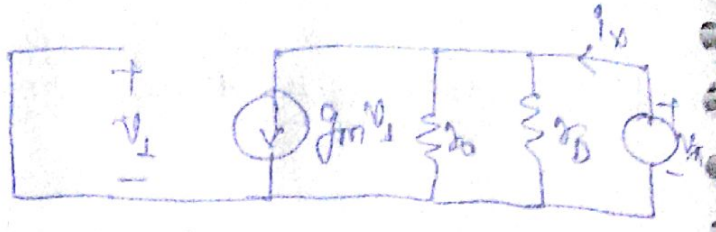
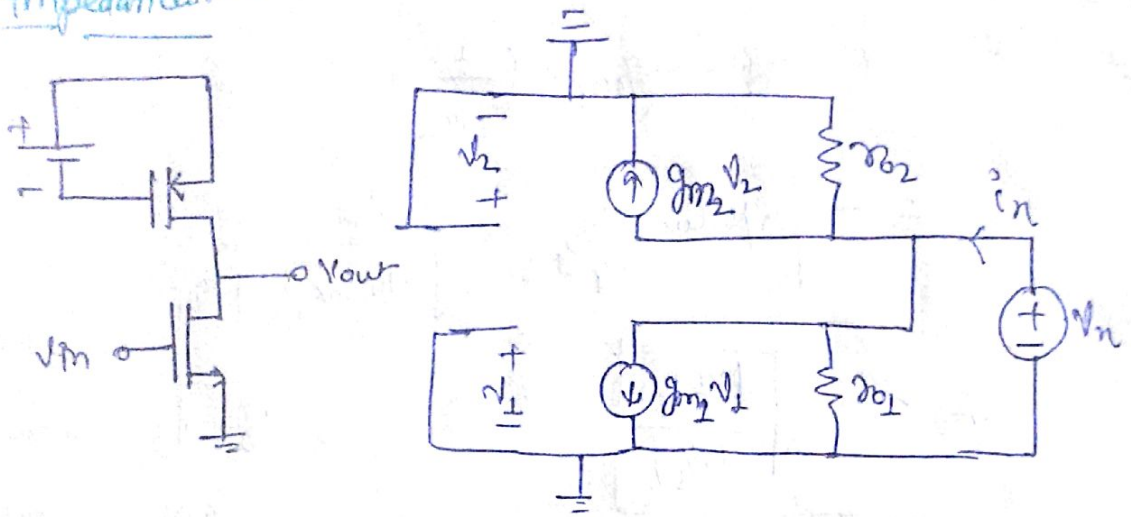




$v_1 = 0$ therefore $g_m v_1 = 0$

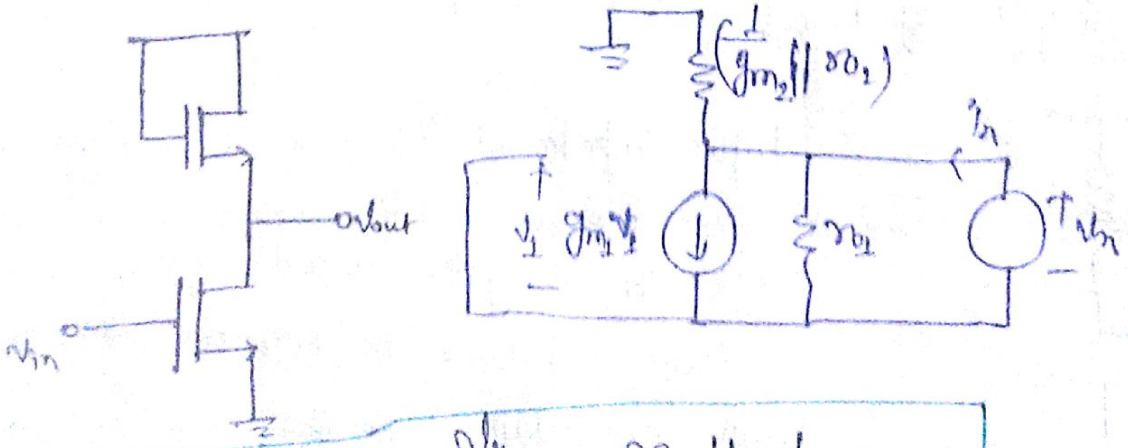
$$\boxed{\text{Output impedance} = \frac{v_x}{i_x} = r_o \| R_D}$$

I/O Impedances:-



$v_1 = v_2 = 0$ so $g_{m_1} v_1 = g_{m_2} v_2 = 0$

$$\boxed{\begin{array}{l}\text{Output impedance} = \dfrac{v_x}{i_x} = (r_{o_1} \| r_{o_2}) \\ \text{Input impedance} = \infty\end{array}}$$



$$\boxed{\text{output impedance} = \frac{v_x}{i_x} = r_{o_1} \| \frac{1}{g_{m_1}} \| r_{o_2}}$$

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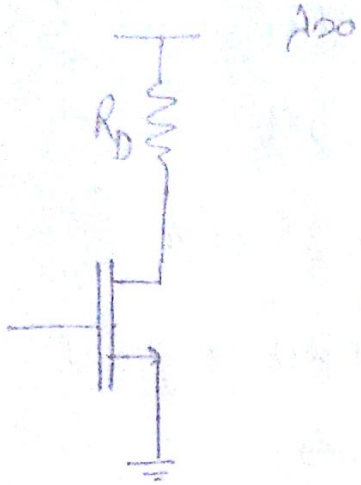
* Problem of Gain Variation:-



$$A_V = -g_m R_D$$

$$\sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}$$

→ factors affecting the gain,

- Temperature
- Supply voltage
- Process
- Signal level

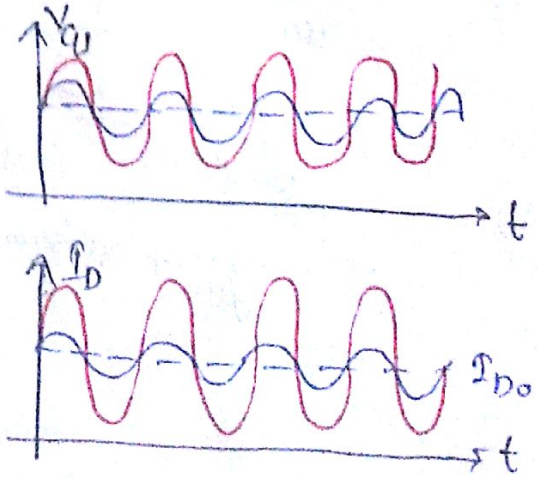
* Mobility is function of temperature so when we go to higher temp so mobility drops therefore g_m drops.

* I_D depends on the supply voltage which may vary. Therefore g_m varies.

* Process variation means the g_m depends on fabrication process which may vary wafer to wafer.

* When we have small signal analysis, change in drain current is negligible with respect to bias value. I_D is relatively constant.

→ If V_{GS} has large signal, I_D has huge fluctuations. Therefore voltage gain also changes with the signal level.



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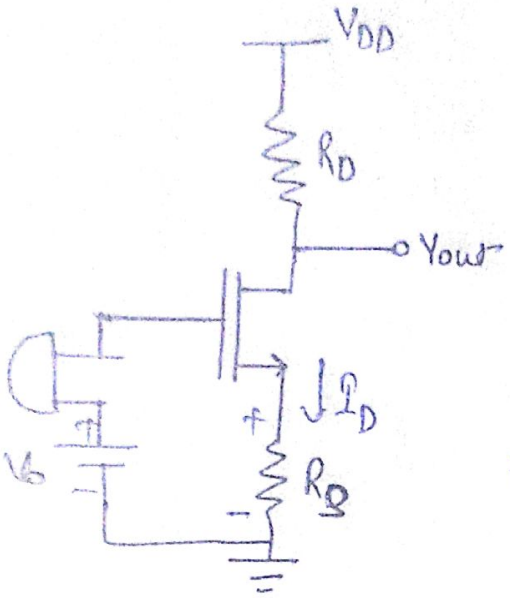
Lecture - 38

• Degenerated CS stage :-

$\lambda = 0$

Step I: Bias conditions



$$V_G = V_{GS} + I_D R_S$$

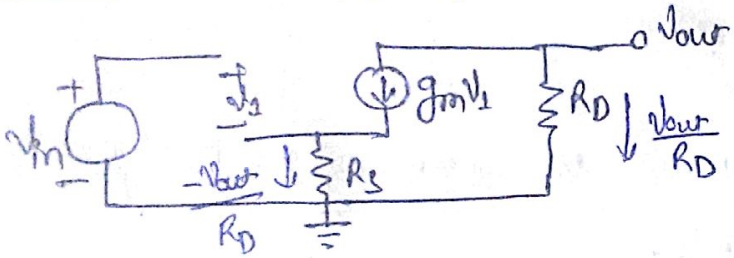
Sat: $I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$

Drain voltage with respect to ground

$$= V_{DD} - I_D R_D$$

For sat: $V_{DD} - I_D R_D \geq V_G - V_{TH}$

Step II: Gain, I/O impedances



$$g_m V_1 = -\frac{V_{out}}{R_D} \Rightarrow V_1 = -\frac{V_{out}}{R_D \cdot g_m}$$

$$V_{in} = -\frac{V_{out}}{g_m R_D} - \frac{V_{out}}{R_D}\cdot R_S$$

$$\Rightarrow \boxed{\frac{V_{out}}{V_{in}} = \frac{-R_D}{\frac{1}{g_m} + R_S}}$$

valid only if $\lambda = 0$

or

$$\boxed{A_v = -\frac{\text{Resistance tied between drain and ac ground}}{\frac{1}{g_m} + \text{resistance tied between source and ac ground}}}$$

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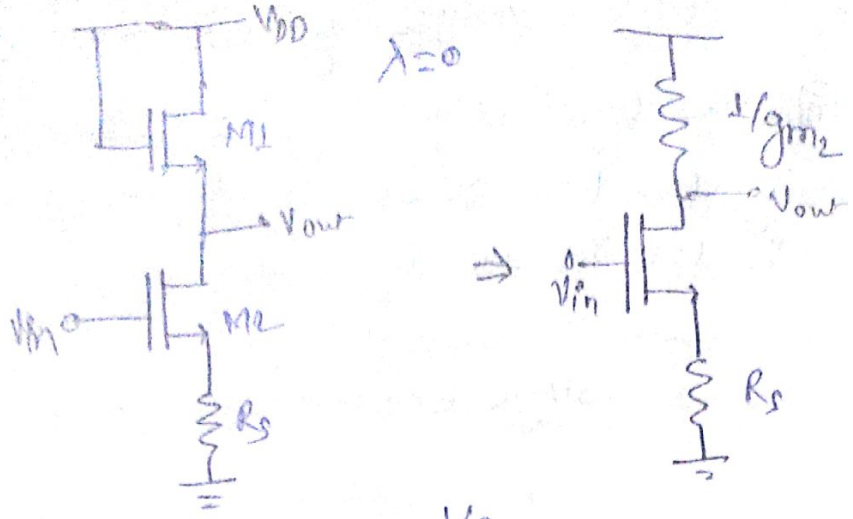
Example



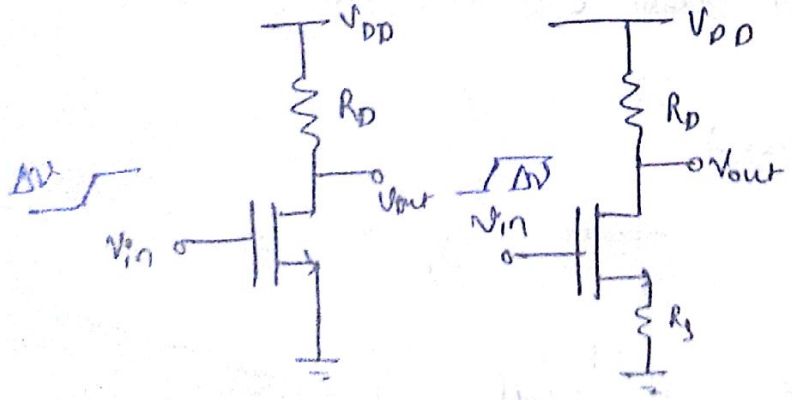
$$A_V = -\frac{1/g_{m2}}{\frac{1}{g_{m1}} + R_S}$$

* In the voltage gain of degenerated CS stage, if g_m varies due to temp., process, supply voltage. If $\frac{1}{g_m} << R_S$ therefore

$$A_V \approx -\frac{R_D}{R_S}$$

* Since circuit becomes less sensitive to gain variation but the gain reduces.

Why is the gain less with degeneration?



$V_{GS} \Rightarrow V_{GS} + \Delta V$

$\Rightarrow \quad I_D \rightarrow I_D + g_m \Delta V$

Assume source voltage is constant even though gate voltage up by ΔV. Therefore I_D will be constant. If V_G goes up by ΔV therefore V_{GS} also goes up by ΔV so current I_D will be increased so source voltage also increases by some amount.

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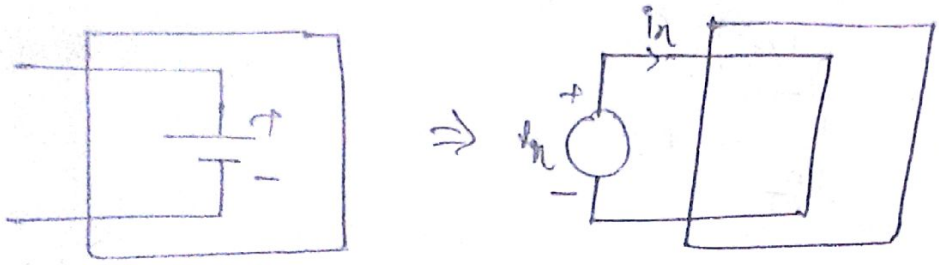
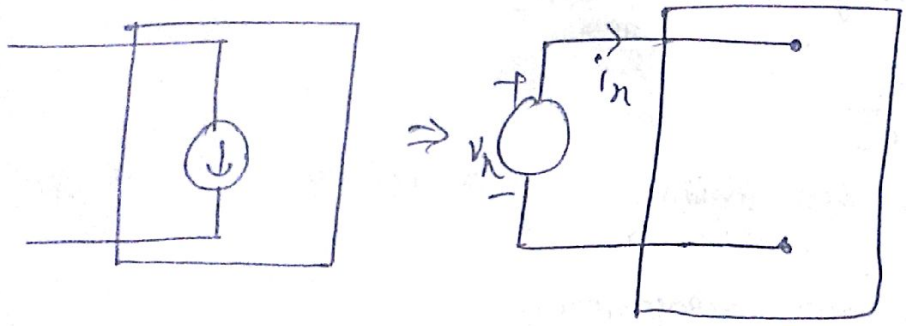
* V_{GS} of degenerated CS stage changes by less than ΔV whereas V_{GS} of simple CS stage changes by ΔV. Therefore current also changes by less than $g_m \Delta V$ in degenerated CS stage. Therefore the gain is less in the degenerate stage.

Impedance of Ideal constant voltage & current sources:-



$\Rightarrow R_{eq} = 0$

→ Small signal impedance of ideal voltage source is zero.

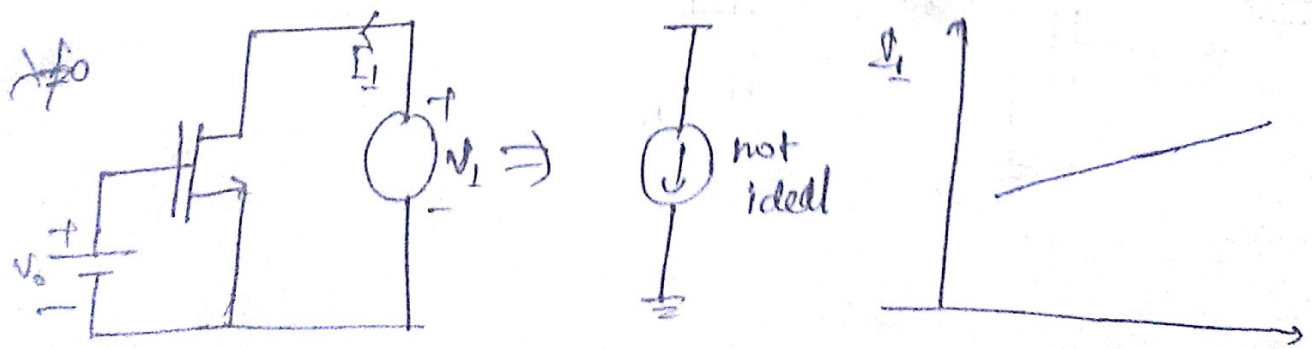


$\Rightarrow R_{eq} = \infty$

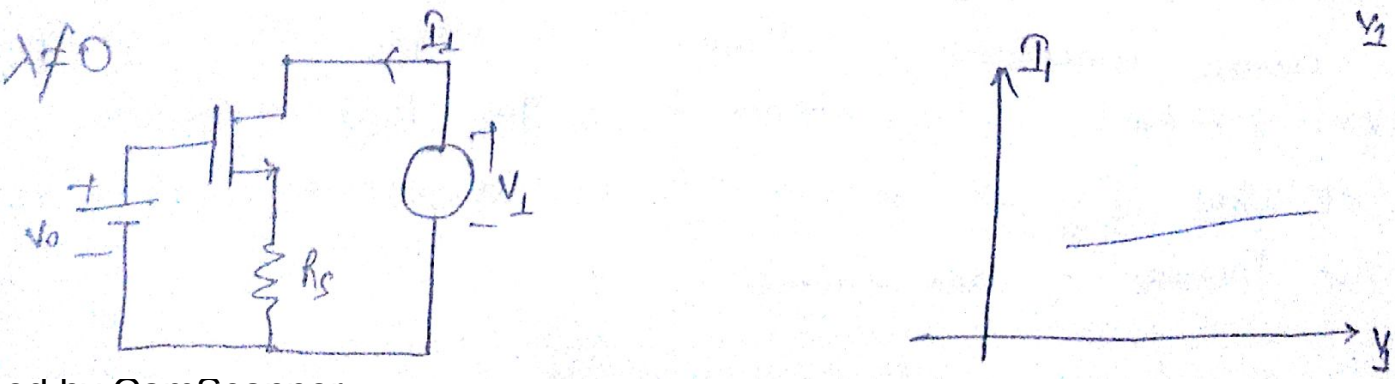
→ Small signal impedance of ideal current source is infinite.

Let's build a current source:-

$\lambda \neq 0$



$\lambda \neq 0$



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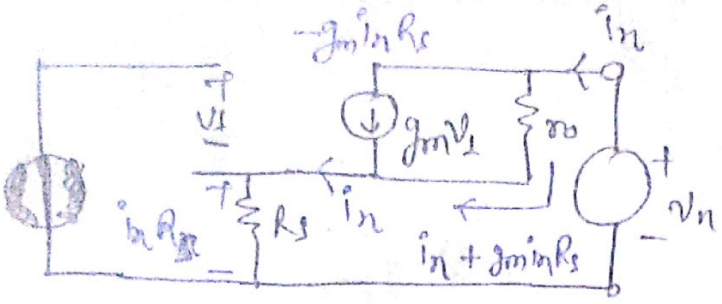
* The degenerated current source is better than the simple current source because it has more flatter curve than that.



$$V_1 = -i_x R_s$$

Current through r_o = $i_x + g_m i_x R_s$

KVL at output port,

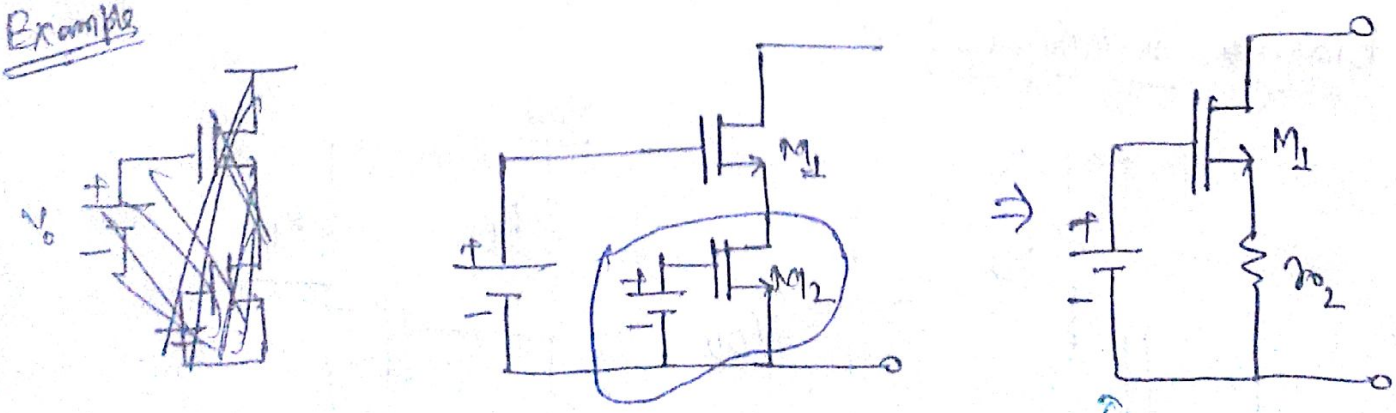
$$V_x = r_o (i_x + g_m R_s i_x) + i_x R_s$$

$$\frac{V_x}{i_x} = r_o(1 + g_m R_s) + R_s$$

Imp. $$\boxed{\frac{V_x}{i_x} = R_s(1 + g_m r_o) + r_o}$$

* As a result of degeneration, output impedance is boosted by $(1 + g_m R_s)$.

Example



cascode structure

$$R_{out} = R_{eq} = (1 + g_{m_1} r_{o_1}) r_{o_2} + r_{o_1}$$

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